

Variapsody: Creating Three Interactive Music Listening Experiences that Use Diversified Positive Emotion Regulation Strategies to Promote Subjective Well-being

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ABSTRACT

Over the past decade, initiatives to design for subjective well-being have gained increased attention and momentum in design research. These initiatives often draw from positive psychology to explore ways of making Positive Psychology Interventions (PPIs) more effective through technology. This paper explores how a mix of tangible and digital technology can realize activity-focused, diverse Emotion Regulation (ER) for its users. We propose that ER strategies can serve as a principle for designing technology that encourages users to savor, modify, reassess, or commemorate their experiences. By centering the design around music listening experiences, the paper explores how users can be supported to lower motivation hurdles that get in the way of frequent engagement with a PPI by developing three exploratory prototypes and reflecting on the development processes. Variapsody, a music listening device, integrates three features, each deploying a different set of ER strategies that make the experience more enjoyable and meaningful. Variapsody's regulatory diversity offers users the choice of how to approach music listening and expands their repertoire of ER strategies. The first feature, Reaction Tile, inscribes users' reactions to music onto a tangible, domino-sized tile to encourage them to savor the music. The second is Monofilter, which purposefully muffles the salience of background music while working on a cognitively demanding task. Vibelist is the third feature that helps users capture and revisit the context of music listening experiences in a digital collage. The paper discusses the lessons learned and future research opportunities.

CCS CONCEPTS

• User Centered Design; • Activity Centered Design; • Contextual Design;

KEYWORDS

Design for well-being, User experience, Emotion regulation, Positive emotion, Positive design, Positive technology, Music listening, Behavioral intervention technologies

ACM Reference Format:

Majed Boughanem and Jungkyoon Yoon. 2022. Variapsody: Creating Three Interactive Music Listening Experiences that Use Diversified Positive Emotion Regulation Strategies to Promote Subjective Well-being. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519743>

1 INTRODUCTION

The exchange of knowledge and methods between positive psychology and design has assisted both fields in pursuing well-being-driven outcomes [24]. On the one hand, positive psychology inspires designers by lending established scientific knowledge about how people flourish and what they consider important in life, contributing to individuals' and communities' well-being. On the other hand, design can support positive psychology's pragmatic goals of developing effective tools and methods for people looking to improve their well-being. In particular, the exchange has advanced the development of Positive Psychology Interventions (PPIs) in the last decade. PPIs are tools devised to nurture users' positive thoughts, behaviors, and feelings [3]. Examples are recurring visits to self-administered apps that offer therapeutic instructions, self-reflection in a journal, or reading self-help books that convey research insights from positive psychology.

While design has contributed to making the evidence-based research insights tangible and actionable for users, some practical challenges remain in how PPIs can be effectively integrated into users' everyday lives. A common barrier to embracing conventional PPIs is their incompatibility with one's lifestyle [32]. For instance, users need to carve out time for therapy and, for those who do not read or write regularly, a journal or self-help book requires a self-driven commitment to forming a new habit. Consequently, PPIs that require users' effortful commitment may be perceived to be obtrusive, hindering their sustained engagement.

In response to these challenges, we propose incorporating PPIs into users' daily activities. Rather than requiring users to add a new task of using PPIs, it may be more compelling and acceptable for them to use such PPIs as part of their daily routines. The frequent nature of daily activities would facilitate recurring opportunities to engage in applying PPIs. Similar to Calm Technology and Slow Computing [20, 21], while this approach would not entirely eliminate the requirement for an intentional effort from users, it may nudge them to integrate the use of PPIs into their everyday contexts. Specifically, the designed outcome of this paper (Figure 1) targets the case of music listening experiences to demonstrate this approach (i.e., PPIs integrated into music listening as part of users'

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519743>

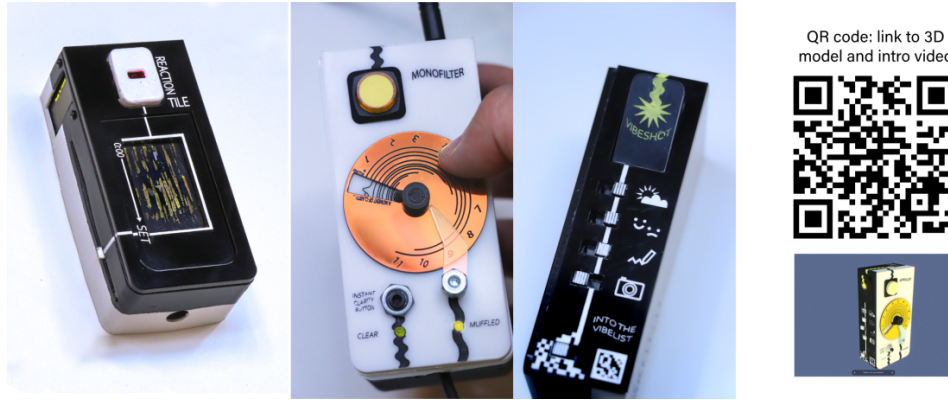


Figure 1: Variapsody’s three features that support users to engage in multiple emotion regulation strategies. Note that while the three features are prototyped separately, the proposed device is one that integrates the three features (each occupying a face of the device; refer to the 3D model linked by the QR code)

Table 1: The five ER strategies according to the process model of ER 11 with explanations and examples.

ER Strategy	Attentional Deployment	Cognitive Change	Situation Modification	Situation Selection	Response Modulation
Description	Directing the attention towards or away from emotional stimuli	An internal modification of judgements, opinions, or perspective	Modifying emotionally influential elements of our surroundings	Choosing to approach desired emotions or to avoid undesired ones	A response to emotions that arise, to suppress or boost them
Situation example	Attending to soothing sounds (e.g., fountain) to ignore unpleasant ones (e.g., noise from the crowd)	Appreciating one’s achievements and avoiding social comparison (e.g., “it will be great”)	Tidying up or adjusting lights to be optimal for listening to music	Choosing to go out for dinner or to an event, leaving a loud venue	Laughing or smiling, saying “I’m so happy.”

daily routines). The following two sections review the theoretical underpinning of the work and additional design considerations for developing the interactive music listening experience: (1) strategies for emotion regulation, and (2) the advantages of tangibility in enhancing human-technology interactions.

2 THEORETICAL BACKGROUND AND DESIGN CONSIDERATIONS

2.1 Designing for emotional regulation

Among the several ways of enhancing well-being, this paper focuses specifically on the design that supports emotion regulation [27]. Emotion Regulation (ER) refers to “how we try to influence which emotions we have, when we have them, and how we experience and express these emotions [11].” A PPI that focuses on ER can have various long-term positive effects on well-being [1] [27]. Despite their fleeting nature, emotions have a long-term impact because they strongly influence the ways people perceive, think, and behave [10]. In particular, positive emotions are known to increase one’s creativity, willingness to try novel activities, and facilitate forming lasting social connections [27]. Therefore, enabling and supporting

the capacity for ER has been considered an effective approach to improving well-being [5]. This paper builds on the five ER strategies drawn from the process model of ER [11], which are described in Table 1 including brief explanations and example scenarios.

People engage in ER in both an implicit and explicit manner and develop their preferences for some strategies over others [4]. Although people regulate their emotions by employing diverse strategies, designs for ER typically address one strategy at a time. For example, Purpal [35], an interactive device that encourages users to anticipate future positive experiences, primarily focuses on “attentional deployment,” while ViBreathe [36], a device that trains respiration techniques with haptic feedback, utilizes “response modulation” as the main ER strategy. However, because different situations require different ER strategies, it would be more advantageous to support users to use diverse ER strategies through technology instead of focusing on one strategy in isolation. For example, attentively listening to music is different from listening to music while working or meeting friends, thereby requiring a different design to make the experience enjoyable and meaningful. In line with this proposition, the designed outcome of this paper

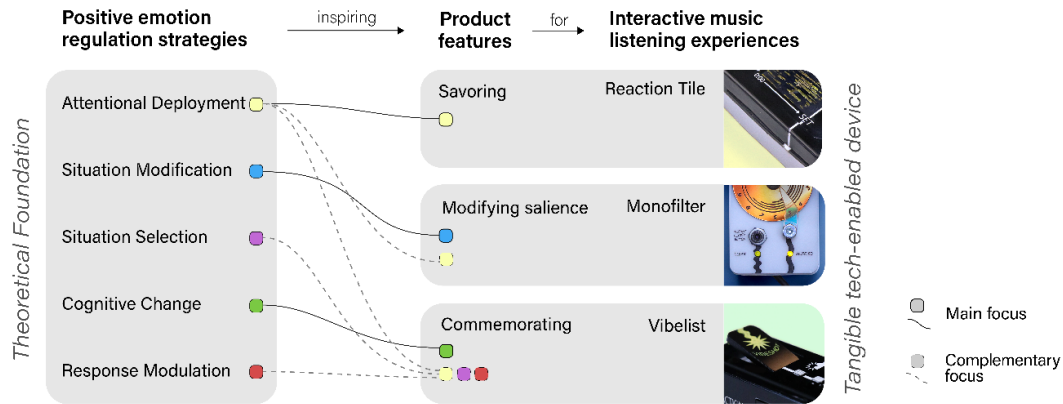


Figure 2: Diagram visualizing how the project is organized conceptually from theory to application

represents multiple ER strategies in the form of three distinct yet complementary interactive music listening features. The goal was to offer people the choice to use the one they find most suitable for their situational needs. Figure 2 illustrates how the three features embody different ER strategies.

2.2 Design consideration: Benefits of tangibility in using activity-focused PPIs

While digital technology-based interventions (such as online PPIs, OPPIs) brings many benefits [2], such as non-consumable interventions that can be repeated, the tangibility of an artifact can add value to making PPIs more engaging to use and elevate their integration into people’s daily contexts [7, 22]. Being physical, tangible PPIs and their outcomes (e.g., receipts or tokens as a symbolic representation of the experience) can serve as visual reminders to use them and help users recall their experiences with the PPI. A dedicated physical artifact can be placed in an appropriate spot that reminds users that they can use it [17] and spontaneously leads to other relevant choices or resources. This aspect can be advantageous if the PPI is placed near other activity-enabling items that can serve as a trigger. This allows the context of the experience to embed itself (directly or symbolically) into the artifact to offer users a salient starting point for the desired experience at hand. In line with these benefits, this paper investigates the possibility of increasing users’ well-being by designing tangible user interfaces that mediate ER.

3 VARIAPSODY: DESIGNING THREE INTERACTIVE MUSIC LISTENING EXPERIENCES

To enhance the experiential benefits for its users, the Variapsody caters to a range of positive music listening experiences. It does so by pairing its main features with the three main ER strategies with appropriate music-listening cases in mind (see Figure 2 for an overview): (1) Reaction Tile, (2) Monofilter, (3) Vibelist. Reaction Tile supports attentional deployment by encouraging users to savor the music in the moment. Monofilter supports situation modification in a way users can effectively balance their cognitive

load while working on a task, listening to music. Vibelist facilitates cognitive change by helping the users interweave the music with other parallel experiential elements (e.g., positivity reinforced by social music listening as demonstrated by Tibau et al. [33, 34]), creating a collage suitable for future reminiscence.

3.1 Reaction Tile

Reaction Tile materializes one’s emotional response to augment the experience of savoring music as a means of facilitating attentional deployment. Reaction Tile visually transcribes the transient emotional response to music onto a domino-sized, rectangular artifact. Operating the reaction tile requires a short setup followed by a period of active listening and finally the taking out of the tile at the end of the experience (See Figure 3). The user inserts a blank tile during its setup and inputs the song duration using the slider on the side. The effort in this phase (i.e., a sunk cost) is meant to allow the user to commit their time to this experience. When ready, the user places their finger on a heart rate monitor and simultaneously starts the music. The engraving tool leaves a trace whose width visually represents their heart rate. The outcome is a tangible, visual representation of one’s emotional unfolding over the song’s duration. When finished, the tile is taken out and stored, collected, shared, etc.

The promise of a tangible reward (i.e., the ‘tile’) is expected to induce a user to deliberately appreciate the music by deploying their genuine and sustained attention to the experience of music listening. Further, salient memories associated with previous listens of the song (e.g., memories of hearing it at a party, or with a partner) may surface. Thus, a tile that captures a pleasurable listening experience serves as a reminder for revisiting the current listening experience [27] in addition to any cognitive component. Reaction Tile tracks emotional valence via physiological response [13, 30], and it may be that a driver of the response is the cognitive component (e.g. reminiscence of an autobiographical memory) that the song being played is associated with [16]. While reminiscence, nostalgia and positive mental time travel is not the primary goal of Reaction Tile, the evoking of memories may open the space for cognitive reappraisal and entail therapeutic benefits [15]. In such cases, the

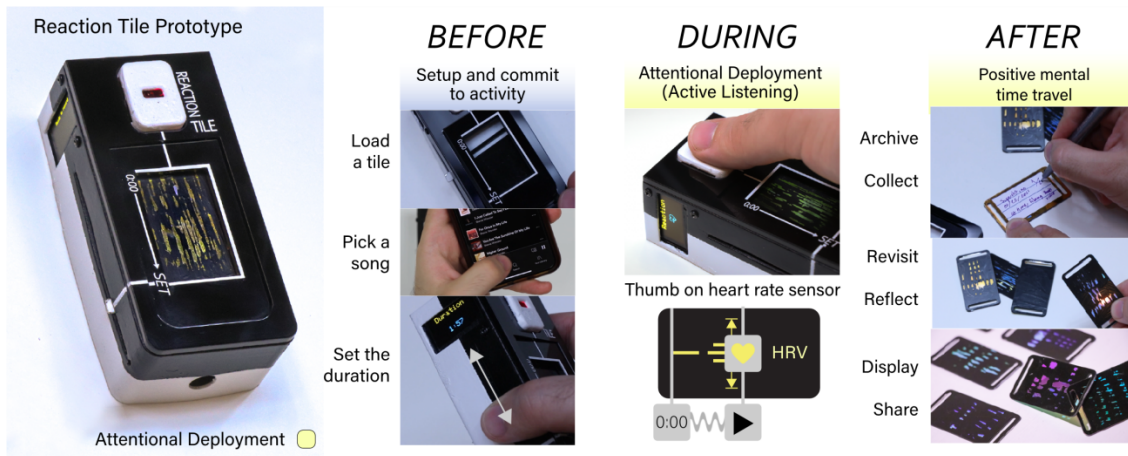


Figure 3: The sequence of how a user uses Reaction Tile. The left-most image is of the prototype that consists of a heart rate sensor at the top, a flap (like cassette players) to insert the tile in. On the side is a slider to set the duration of the song and a small screen to communicate the inputted song duration and confirm the reading of the heartbeat.

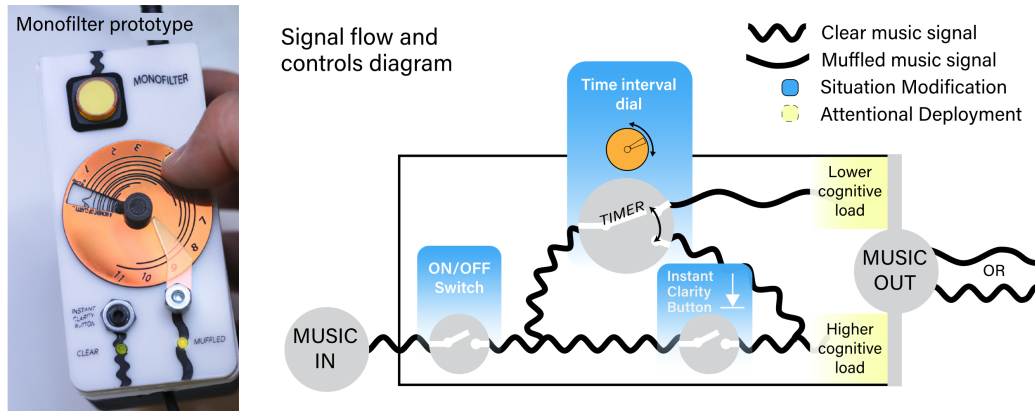


Figure 4: Left: A closeup of Monofilter showing the layout of the controls. Right: The internal working of Monofilter. Note that while the device’s function boils down to a binary outcome when it comes to the music quality (clear/muffled), there are few ways for a user to control the frequency and timing (opportunities for situation modification).

cognitive utility of the tile may incentivize a user to display a reaction tile in their home or turn it into a jewelry item to help frequently revisit or reassess a listening experience.

3.2 Monofilter

Although people enjoy listening to background music in their effort to stay motivated when working on daily activities (e.g., reading a book, following a new recipe, and driving on an unfamiliar path), many of our daily activities require a share of cognitive bandwidth, which makes ‘multi-tasking’ (i.e., the rapid switching of attention between tasks) undesirable for handling the activity. For example, suppose a situation in which someone says, “Turn down the radio. I need to concentrate on writing a report.” In this scenario, highly stimulating music can be too distracting or cognitively burdensome. It was postulated that the overall experience could benefit from controlling the salience of music (beyond just volume control).

Monofilter is a feature for situation modification that offers the ability to take advantage of music as a motivational sidekick to working (see Figure 4). It applies a series of sound effects to the music to reduce the cognitive load that the music can cause. Specifically, Monofilter approaches attentional trade-offs by modulating the clarity of music: muffling/blurring it for a specific duration but periodically revealing it. To operate Monofilter, the user connects their music to the device and selects a “muffling duration” by rotating the central dial. The result is a recurring pattern of muffled music (e.g., two minutes) followed by shorter “moments of clarity” (e.g., five seconds). Through trial and error, a user can find an optimal muffling duration for the activity at hand. If one notices a music section they particularly enjoy, the “instant clarity button” can be clicked to temporarily bypass the filter. This prevents frustration and safeguards a user’s sense of autonomy.

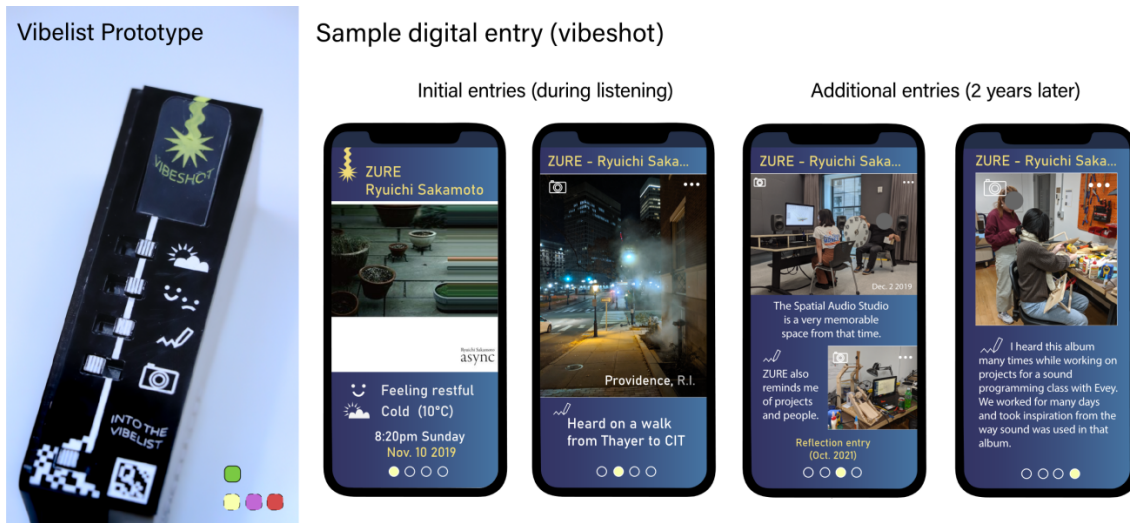


Figure 5: Left: Vibelist feature interface. Before "Vibeshot" is clicked, the switches underneath toggle what to include (weather + location, feeling, text, image and a link for sharing). Right: An example of a vibeshot that guides the user to review the captured information.

3.3 Vibelist

Music is a layer through which we recall, organize, or experience significant life events; songs are associated with places, people, moods, and memories [31]. Vibelist sets the stage for cognitive change by offering ways of documenting, curating, and revisiting the contexts of music listening in retrospect (see Figure 5). The concept of Vibelist was inspired by the studies in psychology that investigated the positive effects of mental time traveling (i.e., an emotion-regulating activity in which an individual vividly remembers or anticipates positive events) on well-being [26]. Vibelist is a feature that enables a user to document and curate "vibeshots"—the equivalent of a screenshot, but of a broad vibe. The constituents of a "vibeshot" are portrayals of external (location, weather, a photo of the surroundings or people) and internal (self-reported feelings and thoughts) elements that are captured in personally meaningful ways.

Vibelist makes the conventional journaling activity music-centric, more engaging, and convenient. Convenience comes from the ease of creating an entry: it requires a decision on the relevant elements and a click of a button. One can type out the text entry and attach photos immediately or after the experience. One can also embed additional contextual information (e.g., weather, location, and currently playing song) as a memory cue for self-reflection. The latter can be obtained automatically (e.g., via sensors and APIs). The option of adding or removing any elements later further enables users to curate the captured narratives as they deem it appropriate.

4 DISCUSSION

This paper explored how a PPI could be integrated into everyday activities, effectively lowering the barrier to using it. Positive emotional experiences derived from music listening is the activity chosen for exploring this design intention. Instead of traditional therapeutic PPIs (e.g., self-help books), we aimed to enable users

to rearrange their familiar daily activities in a way they find more enjoyable and meaningful, thus increasing the chance of increasing their well-being. Inspired by the process model of ER [11], we developed three features of the Variapsody, each supporting activities that upregulate positive emotions based on a different set of ER strategies. The utilization of varying ER strategies was motivated by the proposition that being equipped with diverse ER strategies through the Variapsody (one device offering multiple features) would help users actively adapt to diverse situational opportunities that could potentially facilitate positive music listening experiences.

While functionally separate, we expect that the three features could work in complementary ways. We see it as an added value to consolidate three features into one device. For example, Reaction Tile could enhance the experience of using Vibelist by adding an account of an emotional experience conveyed by the tile to a digital space containing associations between music and other experiential aspects captured in a vibeshot. The sequence of attentional deployment supported by Reaction Tile and cognitive change mediated by Vibelist would reinforce and prolong the positive experiences of music listening. Similarly, Reaction Tile can also make Monofilter more personalized in its stimulation modulation. The primary function of Monofilter is to keep certain musical moments within reach, but not too frequent. Since a reaction tile captures the highly emotional moments, a user could refer to it when adjusting Monofilter (e.g., reverse-transcribing the high moments into moments of clarity while filtering the rest). In this vein, we postulate that users could actively shape their positive experiences by choosing the features in combination or separately, depending on their situational needs and opportunities.

In designing the three features of Variapsody, we tried to ensure their effectiveness by balancing the physical and digital aspects. While digital technology offered several functional advantages (e.g.,

Vibelist's tech-assisted access to surroundings and virtually non-consumable property of a vibeshot), the tangibility of the design opened several experiential opportunities. For example, Reaction Tile created tangible tiles that could serve as souvenirs, allowing for positive mental time travel and savoring the experience after listening to music. It also allowed for sharing and displaying.

By taking music listening as an example experience to design for, we carefully selected three use cases that were deemed appropriate for ER: (1) savoring the music in the moment (supported by Reaction Tile), (2) helping the user balance their cognitive load while working on a task, listening to music (supported by Monofilter), and (3) associating the music with other experiential elements (supported by Vibelist). As illustrated in the three use cases, our experience indicates that design can facilitate ER effectively by supporting users to engage in appropriate activities. Therefore, we suggest that the design process for ER needs to focus on identifying effective and relevant activities in the intended experience and specifying how such activities can be integrated into the interactions with technology-supported PPIs. For example, if the intended experience is hiking, the design process could start by identifying activities that accord with critical values such as achievement, camaraderie, closeness to nature, and exercise. Then, designers could explore what kinds of ER strategies could best support each of the identified activities and how technology could support them.

Moving forward, Variapsody could be further improved in terms of its contribution to well-being by considering additional theoretical lenses. We acknowledge that, in addition to ER, users' subjective well-being can be supported through other design directions, such as addressing the psychological needs of users [14, 24]. For example, the self-determination theory identifies three overarching psychological needs that contribute to well-being: autonomy, competence, and relatedness [29]. Several meta-analyses (e.g., [6] and [19]) showed that these psychological needs robustly predict increased well-being across various domains (e.g., health, education, and sport). Therefore, it would be worthwhile to examine if and how the use of Variapsody fulfills these needs. For example, it may be beneficial for Monofilter to have relevant user-inputs in place to allow users to take more credit (i.e., autonomy) for succeeding in optimally tuning the cognitive demands of music listening (i.e., competence).

At present, we are interested in learning about how users would use such a device. A promising approach for evaluating technological probes such as Variapsody is to use anticipatory ethnography [28]. By framing it as a fictitious product, user groups could focus on the experiential potential rather than be distracted by details regarding feasibility or realism. To evaluate the actual effects of Variapsody, an in-situ field experiment would be conducted in which participants use the three features in their daily contexts, which will generate new insights regarding how users would integrate ER-enabling technologies into their everyday lives and how the technologies could foster their well-being. In particular, we plan to collect both qualitative and quantitative data regarding users' emotional states, psychological needs, and well-being by using validated measures such as Emotion Regulation Questionnaire [12], Savoring Beliefs Inventory [4], Technology-based Experience of Need Satisfaction [25]. Well-being will be measured at three levels:

(1) Scale of Positive and Negative Experience [9] for emotional well-being, (2) Satisfaction with Life Scale [23] for life satisfaction, and (3) Subjective Happiness Scale [18] for overall happiness. Along with the quantitative measures, qualitative data focused on identifying the reasons and factors that underlie the users' responses to the measures will be collected. Based on the findings, we will refine the current design of Variapsody and continue to develop new technology-supported PPIs and explore detailed application possibilities in different domains (i.e., non-music listening experience), incorporating the lessons gained through developing and testing Variapsody.

5 CONCLUSION

In the past few decades, research in design for well-being [8, 23] has made impressive progress in exploring how findings from positive psychology can inform the process of developing well-being-enhancing technology. In this paper, we explored the possibilities of how technology that supports ER can fit into everyday activities. Specifically, the design takes the form of three interactive music listening experiences, each focusing on a different ER strategy: (1) Reaction Tile prompts attentional deployment during and after savoring music, (2) Monofilter enables situation modification of background music while working on a task, and (3) Vibelist supports the recording and revisiting of music listening contexts. We expect that equipping users with such multi-faceted ER strategies can benefit them in their endeavor to increase their well-being. We hope that this paper will serve as a good starting point for further investigation into the design of technology-supported PPIs and their impact on well-being, and the development of design approaches that assist designers in making a positive impact through their designs.

ACKNOWLEDGMENTS

The research was supported by Cornell Center for Social Sciences. We extend our gratitude to the anonymous peer reviewers for their helpful comments on an earlier version of the paper.

REFERENCES

- [1] Allison A Appleton and Laura D Kubzansky. 2014. Emotion regulation and cardiovascular disease risk. *Handbook of emotion regulation* (2nd ed.), 596-612. The Guilford Press.
- [2] Linda Bolier and Katherina Martin Abello. 2014. Online Positive Psychological Interventions: State of the Art and Future Directions. In *The Wiley Blackwell Handbook of Positive Psychological Interventions*, 286 - 309. Wiley Blackwell. <https://doi.org/10.1002/9781118315927.ch16>
- [3] Linda Bolier, Merel Haverman, Gerben J. Westerhof, Heleen Riper, Filip Smit, and Ernst Bohlmeijer. 2013. Positive psychology interventions: A meta-analysis of randomized controlled studies. *BMC Public Health* 13, 1. <https://doi.org/10.1186/1471-2458-13-119>
- [4] Fred B. Bryant. 2003. Savoring Beliefs Inventory (SBI): A scale for measuring beliefs about savouring. *Journal of Mental Health* 12, 2, 175-196. <https://doi.org/10.1080/0963823031000103489>
- [5] Fred B. Bryant and Joseph Veroff. 2017. *Savoring: A new model of positive experience*. Routledge <https://doi.org/10.4324/9781315088426>
- [6] Nikos L.D. Chatzisarantis, Martin S. Hagger, Stuart J.H. Biddle, Brett Smith, and John C.K. Wang. 2003. A Meta-Analysis of Perceived Locus of Causality in Exercise, Sport, and Physical Education Contexts. *Journal of Sport and Exercise Psychology* 25, 3, 284-306. <https://doi.org/10.1123/jsep.25.3.284>
- [7] Pieter M. A. Desmet and Maria C. Sääksjärvi. 2016. Form Matters: Design Creativity in Positive Psychological Interventions. *Psychology of Well-Being* 6, 1, 1-17. <https://doi.org/10.1186/s13612-016-0043-5>

- [8] Pieter M.A. Desmet and Anna E. Pohlmeier. 2013. Positive design: An introduction to design for subjective well-being. *International Journal of Design* 7, 3, 1-15.
- [9] Ed Diener, Derrick Wirtz, William Tov, Chu Kim-Prieto, Dong won Choi, Shigehiro Oishi, and Robert Biswas-Diener. 2010. New well-being measures: Short scales to assess flourishing and positive and negative feelings. *Social Indicators Research* 97, 2, 143-156. <https://doi.org/10.1007/s11205-009-9493-y>
- [10] Barbara L. Fredrickson. 2013. Positive Emotions Broaden and Build. In *Advances in Experimental Social Psychology*. <https://doi.org/10.1016/B978-0-12-407236-7.00001-2>
- [11] James J. Gross. 1998. The emerging field of emotion regulation: An integrative review. *Review of General Psychology* 2. <https://doi.org/10.1037/1089-2680.2.3.271>
- [12] James J. Gross and Oliver P. John. 2003. Individual Differences in Two Emotion Regulation Processes: Implications for Affect, Relationships, and Well-Being. *Journal of Personality and Social Psychology* 85, 2, 348-362. <https://doi.org/10.1037/0022-3514.85.2.348>
- [13] Luke Harrison and Psyche Loui. 2014. Thrills, chills, frissons, and skin orgasms: Toward an integrative model of transcendent psychophysiological experiences in music. *Frontiers in Psychology* 5, JUL. <https://doi.org/10.3389/fpsyg.2014.00790>
- [14] Marc Hassenzahl. 2010. Experience Design: Technology for All the Right Reasons. *Synthesis Lectures on Human-Centered Informatics* 3, 1. <https://doi.org/10.2200/s00261ed1v01y201003hci008>
- [15] Lauren Istvandy. 2017. Combining music and reminiscence therapy interventions for wellbeing in elderly populations: A systematic review. *Complementary Therapies in Clinical Practice* 28, 18-25. <https://doi.org/10.1016/j.ctcp.2017.03.003>
- [16] Petr Janata, Stefan T. Tomic, and Sonja K. Rakowski. 2007. Characterisation of music-evoked autobiographical memories. *Memory* 15, 8, 845-860. <https://doi.org/10.1080/09658210701734593>
- [17] Matthias Laschke, Sarah Diefenbach, Thies Schneider, and Marc Hassenzahl. 2014. Keypoint: initiating behavior change through friendly friction. In *Proceedings of the 8th Nordic Conference on Human-Computer Interaction: Fun, Fast, Foundational (NordiCHI '14)*. Association for Computing Machinery, New York, NY, USA, 853-858. DOI:<https://doi.org/10.1145/2639189.2670179>
- [18] Sonja Lyubomirsky and Heidi S. Lepper. 1999. A measure of subjective happiness: Preliminary reliability and construct validation. *Social Indicators Research* 46, 2, 137-155. <https://doi.org/10.1023/A:1006824100041>
- [19] Johan Y.Y. Ng, Nikos Ntoumanis, Cecilie Thøgersen-Ntoumani, Edward L. Deci, Richard M. Ryan, Joan L. Duda, and Geoffrey C. Williams. 2012. Self-Determination Theory Applied to Health Contexts: A Meta-Analysis. *Perspectives on Psychological Science* 7, 4, 325-340. <https://doi.org/10.1177/1745691612447309>
- [20] William Odom, Richard Banks, Abigail Durrant, David Kirk, and James Pierce. 2012. Slow technology: critical reflection and future directions. In *Proceedings of the Designing Interactive Systems Conference (DIS '12)*. Association for Computing Machinery, New York, NY, USA, 816-817. DOI:<https://doi.org/10.1145/2317956.2318088>
- [21] William T. Odom, Abigail J. Sellen, Richard Banks, David S. Kirk, Tim Regan, Mark Selby, Jodi L. Forlizzi, and John Zimmerman. 2014. Designing for slowness, anticipation and re-visitation: a long term field study of the photobox. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1961-1970. DOI:<https://doi.org/10.1145/2556288.2557178>
- [22] Acacia C. Parks, Matthew D. della Porta, Russell S. Pierce, Ran Zilca, and Sonja Lyubomirsky. 2012. Pursuing happiness in everyday life: The characteristics and behaviors of online happiness seekers. *Emotion* 12, 6, 1222. <https://doi.org/10.1037/a0028587>
- [23] William Pavot, E. Diener, C. Randall Colvin, and Ed Sandvik. 1991. Further Validation of the Satisfaction With Life Scale; Evidence for the Cross-Method Convergence of Well-Being Measures. *Journal of Personality Assessment* 57, 1, 149-161. https://doi.org/10.1207/s15327752jpa5701_17
- [24] Ann Petermans and Rebecca Cain (eds.). 2019. *Design for wellbeing: An applied approach*. Routledge, New York. ISBN 9781032176529
- [25] Dorian Peters, Rafael A. Calvo, and Richard M. Ryan. 2018. Designing for motivation, engagement and wellbeing in digital experience. *Frontiers in Psychology* 9, MAY, 1-15. <https://doi.org/10.3389/fpsyg.2018.00797>
- [26] Jordi Quoidbach, Elizabeth v. Berry, Michel Hansenne, and Moïra Mikolajczak. 2010. Positive emotion regulation and well-being: Comparing the impact of eight savoring and dampening strategies. *Personality and Individual Differences* 49, 5. <https://doi.org/10.1016/j.paid.2010.03.048>
- [27] Jordi Quoidbach, Moïra Mikolajczak, and James J. Gross. 2015. Positive interventions: An emotion regulation perspective. *Psychological Bulletin* 141, 3. <https://doi.org/10.1037/a0038648>
- [28] Ronda Ringfort-Felner, Matthias Laschke, Shadan Sadeghian, and Marc Hassenzahl. 2022. Kiro: A Design Fiction to Explore Social Conversation with Voice Assistants. *Proc. ACM Hum.-Comput. Interact.* 6, GROUP, Article 33 (January 2022), 21 pages. DOI:<https://doi.org/10.1145/3492852>
- [29] Richard M. Ryan and Edward L. Deci. 2000. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist* 55, 1. <https://doi.org/10.1037//0003-066x.55.1.68>
- [30] Valorie N. Salimpoor, Mitchel Benovoy, Gregory Longo, Jeremy R. Cooperstock, and Robert J. Zatorre. 2009. The rewarding aspects of music listening are related to degree of emotional arousal. *PLoS ONE* 4, 10. <https://doi.org/10.1371/journal.pone.0007487>
- [31] Thomas Schäfer, Peter Sedlmeier, Christine Städtler, and David Huron. 2013. The psychological functions of music listening. *Frontiers in Psychology* 4, AUG. <https://doi.org/10.3389/fpsyg.2013.00511>
- [32] Stephen M. Schueller and Acacia C. Parks. 2014. The Science of Self-Help. *European Psychologist* 19, 2. <https://doi.org/10.1027/1016-9040/a000181>
- [33] Javier Tibau, Michael Stewart, Steve Harrison, and Deborah Tatar. 2019. FamilySong: Designing to Enable Music for Connection and Culture in Internationally Distributed Families. In *Proceedings of the 2019 on Designing Interactive Systems Conference (DIS '19)*. Association for Computing Machinery, New York, NY, USA, 785-798. DOI:<https://doi.org/10.1145/3322276.3322279>
- [34] Javier Tibau, Michael Stewart, Steve Harrison, and Deborah Tatar. 2019. FamilySong: A Design for Managing Synchronous Intergenerational Remote Music Sharing. In *Companion Publication of the 2019 on Designing Interactive Systems Conference 2019 Companion (DIS '19 Companion)*. Association for Computing Machinery, New York, NY, USA, 61-64. DOI:<https://doi.org/10.1145/3301019.3325159>
- [35] Jung Kyoon Yoon, Shuran Li, and Yu Hao. 2021. Design-Mediated Positive Emotion Regulation: The Development of an Interactive Device that Supports Daily Practice of Positive Mental Time Traveling. *International Journal of Human-Computer Interaction*. <https://doi.org/10.1080/10447318.2021.1948685>
- [36] Bin Yu, Pengcheng An, Sjoerd Hendriks, Ning Zhang, Loe Feijs, Min Li, and Jun Hu. 2021. ViBreathe: Heart Rate Variability Enhanced Respiration Training for Workaday Stress Management via an Eyes-Free Tangible Interface. *International Journal of Human-Computer Interaction* 37, 16. <https://doi.org/10.1080/10447318.2021.1898827>